

Naan Mudhalvan - Project

Topic Name	Importing & Securing Data in ServiceNow
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Department	Computer Science and Engineering (Cyber Security)
Year / Sem	III - Year / V - Sem
College Name	Jaya Sakthi Engineering College

Importing & Securing Data in ServiceNow

Import Sets, Dot-Walking and Access Control Lists (ACL)

1. Description

Problem Statement

Each training record must be linked to an employee, and employee details such as department must be pulled into the record for easier reporting.

Objective

To implement an Access Control List (ACL) and verify it using Impersonate User, giving a solid foundation for managing and securing data in ServiceNow using key platform features.

2. Process

Milestone 1: Tables (Create Table)

1. Open ServiceNow, go to All and search for Tables, then select Tables under System Definition and click New.
2. Enter Label: Employee Training Records and Name: u_employee_training_records; tick Create module and Add module to menu.
3. Add columns: Training Name (String), Completion Date (Date), Status (Choice), Employee (Reference to sys_user) and click Submit.
4. Open the Status field and add choices: Inprogress (ip) and Completed (c).

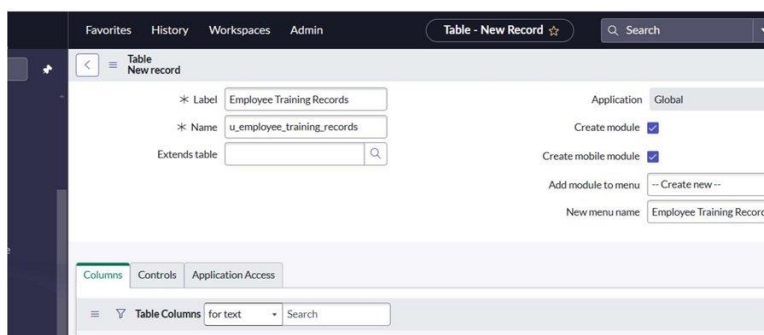
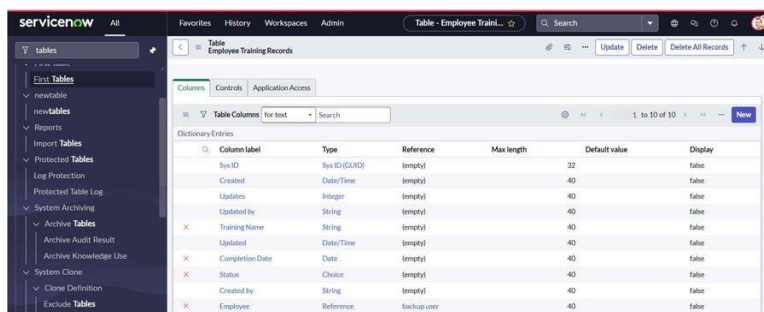


Fig 1: Create table



Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32	false	false
Created	DateTime	(empty)	40	false	false
Updated by	Integer	(empty)	40	false	false
Updated	String	(empty)	40	false	false
Training Name	String	(empty)	40	false	false
Updated	DateTime	(empty)	40	false	false
Completion Date	Date	(empty)	40	false	false
Status	Choice	(empty)	40	false	false
Created by	String	(empty)	40	false	false
Employee	Reference	backup user	40	false	false

Fig 2: Table columns

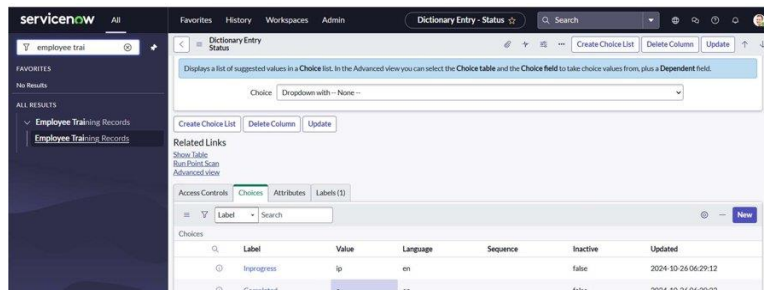


Fig 3: Status choices

Milestone 2: Import Data and Map Fields

1. Create an Employee Training .xlsx sheet with Training Name, Completion Date, Status and Employee, and download it.
2. Go to All > System Import Sets > Load Data, upload the file with Label: Employee Training and Name: u_employee_training, then Submit.
3. Go to System Import Sets > Administration > Transform Maps and create a map from the Employee Training import set to the Employee Training Records table.
4. Add field maps for the four fields and click Transform under Related Links to run the import.

	A	B	C	D	E	F	G	H	I
1	u_training_name	u_completion_date	u_status	u_employee					
2	ServiceNow Fun	2026-01-15	Completed	System Administrator					
3	ITSM Fundamer	2026-01-22	Completed	Abel Tuter					
4	ServiceNow Scri	2026-02-05	In Progress	Beth Anglin					
5	Flow Designer B	2026-02-12	Completed	David Loo					
6	Client Scripts an	2026-02-20	Completed	Fred Luddy					
7	ACL and Securit	2026-03-03	In Progress	ITIL User					
8	Business Rules	2026-03-10	Not Started	Joe Employee					
9	Service Catalog	2026-03-18	Completed	System Administrator					
10	Integration Fund	2026-03-25	In Progress	Abel Tuter					
11	REST API Funda	2026-04-02	Completed	Beth Anglin					
12	GlideRecord Tra	2026-04-10	Completed	David Loo					
13	Script Includes	2026-04-17	Not Started	Fred Luddy					

Fig 4: Source spreadsheet

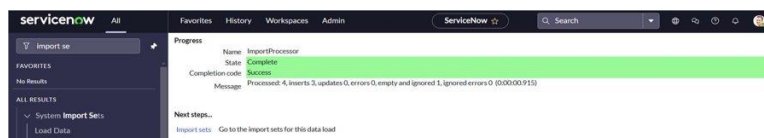


Fig 5: Data imported

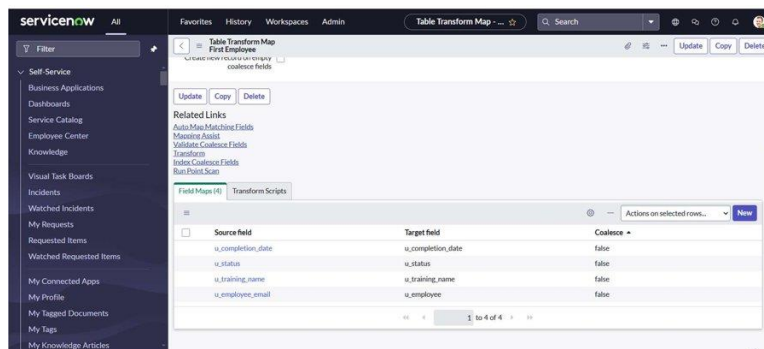


Fig 6: Field maps

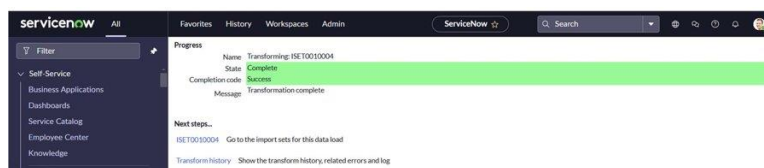


Fig 7: Transform complete

Milestone 3: Dot-Walking to Access Employee Department

1. Open Employee Training Records and click New.
2. Select Additional Actions > Configure > Form Layout.
3. Expand Employee [+], dot-walk to Department, move it to the Selected list and Save.

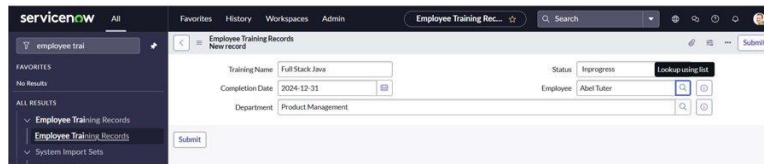


Fig 8: Department via dot-walking

Milestone 4: Roles and User Creation

1. Create a user named backup user (All > User Administration > Users).
2. Create a role named Hr Manager (User Administration > Roles).
3. Open backup user, edit Roles and add the Hr Manager role.
4. Add the Hr Manager role to the Employee Training Records application menu and its module.

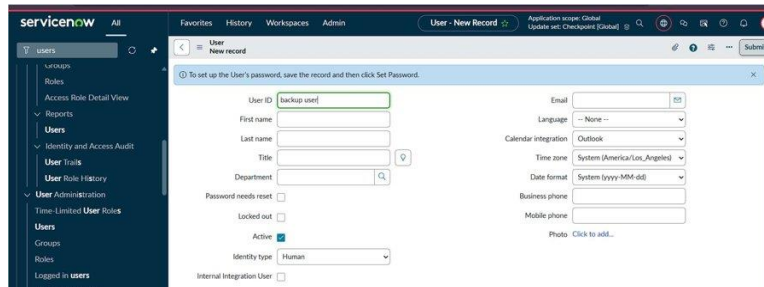


Fig 9: Create user

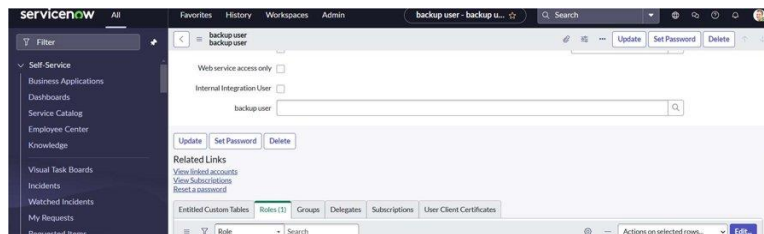


Fig 10: Role assigned

Milestone 5: Access Control List (ACL)

1. Click Profile > Elevate Role, tick security_admin and Update.
2. Create a new ACL (Type: record, Operation: read) on u_employee_training_records and add the Hr Manager role.
3. Create another ACL with Operation: write on the same table and add the Hr Manager role.

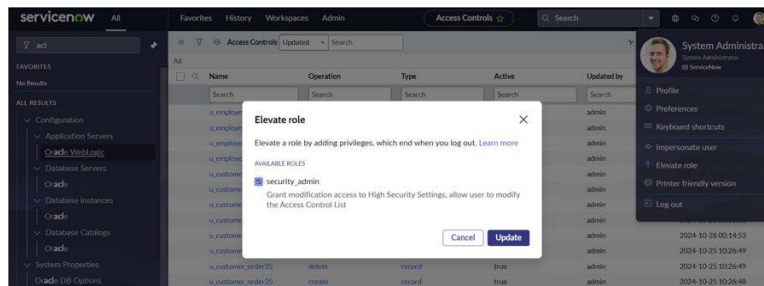


Fig 11: Elevate role

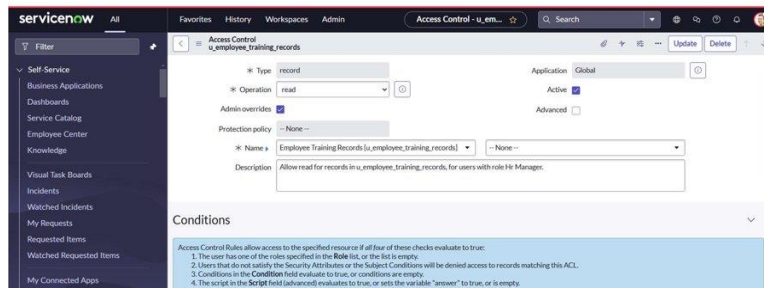


Fig 12: Read ACL

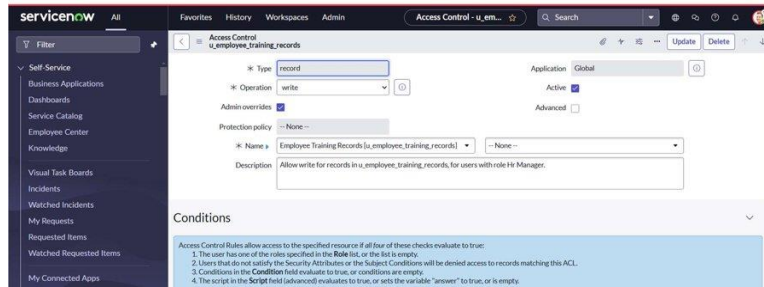


Fig 13: Write ACL

Milestone 6: Testing with Impersonation

1. Impersonate the backup user and search Employee Training Records.
2. The records can be viewed and edited, and the Department field is displayed.
3. Impersonate any other user and the table is not visible.

Completion Date	Employee	Status	Training Name
2024-10-27	Albert Tubler	In progress	Full stack java
2024-12-21	Albert Tubler	In progress	Full Stack Java

Fig 14: Records visible to Hr Manager

Fig 15: Form with Department field

3. Result

Data from the spreadsheet was imported successfully into the Employee Training Records table, and each record was linked to an employee through the Employee reference field.

Using dot-walking, the employee's Department is shown directly on the training record, which makes reporting easier.

The read and write ACLs allow only users with the Hr Manager role to view and edit the table. When the backup user was impersonated, the records could be seen and edited; when another user was impersonated, the table was not visible.

Conclusion: Import Sets, dot-walking and ACLs together ensured accurate data integration, streamlined data relationships and role-based protection of sensitive information.